GreenShift data cards

One card per dataset: licence, access, provenance and known issues. The first three are used; Electricity Maps was substituted by NESO data (see its card).

Data card: GB Carbon Intensity API

Field	Value
Name	Carbon Intensity API (Great Britain)
Publisher	NESO (National Energy System Operator), with Environmental Defense Fund Europe, University of Oxford, WWF
URL	https://api.carbonintensity.org.uk, docs https://carbon-intensity.github.io/api-definitions/
Licence	Creative Commons Attribution 4.0 (CC BY 4.0) + API Terms of Use
Access	Public REST, no account or key
Accessed	2026-10-27
Period used	2024-01-01 → 2025-12-31 (UTC)
Granularity	30-minute settlement periods
Spatial scope	GB national + 14 DNO regions (ids 1-14)
Variables	`forecast`, `actual` (national only), `index`, regional `generationmix` % by fuel
Units	gCO2/kWh (average, production-based, direct combustion emissions)
Personal data	None (grid-level aggregates)
Known issues	Missing `forecast` in some national periods (26/673 rows in 1-14 Jan 2025); a few missing regional half-hours; values may be revised
Pre-processing	De-duplicate on `from`, reindex to a full 30-min UTC grid, interpolate gaps ≤ 2 h, flag longer gaps
Use in project	Scheduler backtest (forecast = decision signal, actual = ground truth for accounting) and the live forecast for the online mode
Attribution	"Carbon intensity data: Carbon Intensity API, NESO (CC BY 4.0)."

Data card: NESO Historic GB generation mix & carbon intensity

Field	Value
Publisher	NESO Data Portal
URL	https://www.neso.energy/data-portal/historic-generation-mix
Licence	CC BY 4.0 according to the dataset page (each NESO dataset is licensed individually, so re-confirm)
Access	Public CSV download, no account
Coverage	2009-01-01 → present, GB, half-hourly
Variables	`DATETIME` (UTC), generation by fuel (MW and %), `CARBON_INTENSITY` (gCO2/kWh), low/zero-carbon, renewable, fossil aggregates
Personal data	None
Known issues	Values revised during cleansing. Some third-party commentary questions the aggregate % columns (storage/solar netting)
Use in project	Cross-check of the API `actual` series, and a long-run trend of shifting potential
Provenance	File SHA-256 recorded at download (`scripts/fetch_carbon_data.py` output)

Data card: TPC-DS-derived workload

Field	Value
Source	TPC Benchmark DS (spec v4.0.0), Transaction Processing Performance Council, https://www.tpc.org
Generator used	DuckDB 2.0.0 `tpcds` extension (`CALL dsdgen(sf)`, `tpcds_queries()`), a port of the TPC-DS v4 tools
Licence	TPC EULA + TPC Fair Use Policy. Results are **not** audited TPC-DS results and must not be presented as such
Data	Synthetic retail data warehouse: 24 tables (7 fact, 17 dimension). No real people, so no re-identification risk (names/addresses are generated)
Scale factors	SF1 (0.28 GB parquet; store_sales 2,880,404 rows). All 99 queries were timed at SF5, but sustained SF5 campaigns exhausted the Java heap, so the measurement catalogue uses SF1 with each query group repeated (`configs/jobs.json`)
Format	Parquet, zstd, one file per table (`data/tpcds/sf{N}/`), manifest.json with row counts and sizes
Queries	99 qualification queries (`data/tpcds/queries/qNN.sql`). Double-quoted aliases converted to backticks for Spark SQL
Use in project	Representative nightly ETL/BI Spark jobs whose energy we measure
Required wording	"Workload derived from TPC-DS; not comparable with published TPC-DS results."

Data card: Electricity Maps (status: optional / pending access)

Field	Value
Publisher	Electricity Maps ApS
URL	https://www.electricitymaps.com (datasets: app.electricitymaps.com/datasets)
Licence	Historical CSV datasets: Open Database License (ODbL, attribution + share-alike). Free API: non-commercial use only
Access	Free account required. As of 2026 the CSV datasets product is reported as **sunset for general users** and available to approved academic users. Free API key gives latest + past-24 h only
Variables	Hourly carbon intensity (direct and life-cycle, consumption-based, flow-traced), carbon-free %, renewable %
Zones of interest	`GB`; India `IN-SO` (South: Bengaluru), `IN-WE`, `IN-NO`, `IN-EA`, `IN-NE` (verify codes on the portal)
Use in project	**Not used yet.** If access is granted: (a) compare the GB life-cycle vs direct signal, (b) add India South as a second, higher-carbon region
If not used	Substituted by NESO data (same system operator, CC BY 4.0, 30-min grain, no key). The justification goes in the report
Share-alike note	If we publish derived Electricity Maps datasets, they must be released under ODbL